Project Overpass

An Agentic Assistant

July 14, 2026

Python · FastAPI · Streamlit · sentence-transformers · ChromaDB · GPT-5-mini

Overview

To build a digital assistant that helps people work through everyday emotional states.
The assistant needs to:

- Recognise a crisis
- Match the feeling to expert guidance
- Respond in a way that is genuinely helpful and, above all, safe
- Draw on philosophical and therapy texts for deeper, open-ended feelings
- Never improvise clinical advice
- Recognise when an input needs a human rather than an automated response

Approach

- Mixed data — three fundamentally different sources needing different handling
- Safety first — escalation must intercept before any response is formed
- Smart routing — not every message needs the same tool
- Grounded responses — never improvise; always cite what you drew on
- Robustness — invalid input, LLM failures, timeouts handled gracefully

Sources — Three Different sets

STRUCTURED

knowledge_base.json

10 emotional states linked via explicit IDs to frameworks, practices, and inquiry / mirror patterns.

Relationships are authored and exact. Exact keyed lookup, not similarity search.

SEED SET

classifier_seeds.json

12 labelled utterances mapping text to state_id + intensity + safety flag.

3 safety levels: none, watch, escalate.

Bootstrap router via embedding similarity to the expandable seed sentences.

UNSTRUCTURED

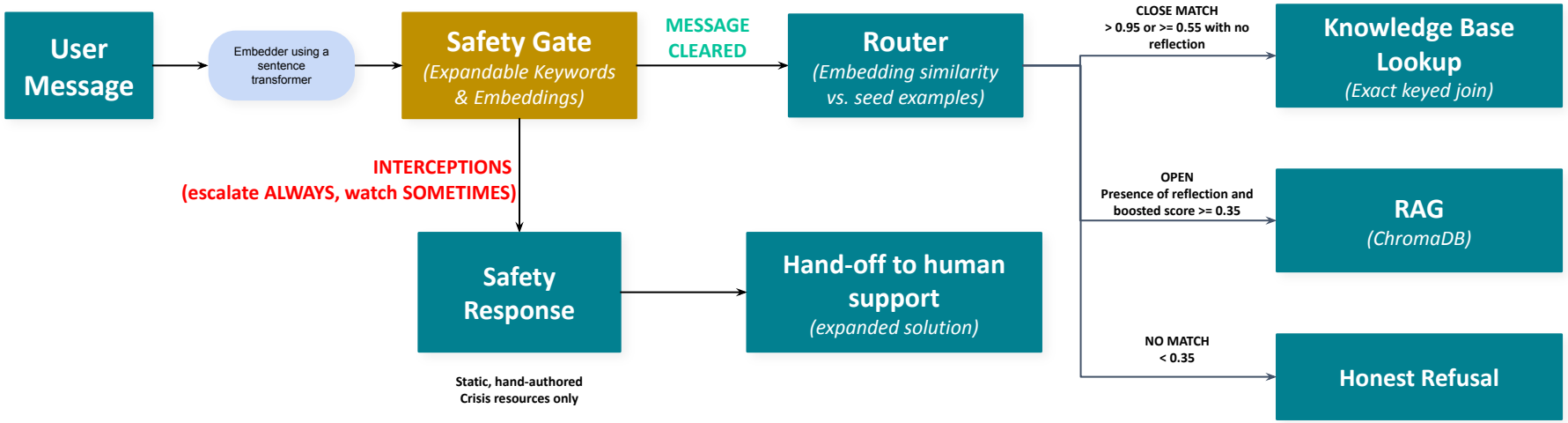
Wisdom Corpus (4 URLs)

multiple multi-format sources — Taoist, Hindu, CBT, ACT traditions.

No join keys. Meaning is everything. Retrieval ⟶ Augmentation ⟶ Generation

Downloaded and processed into small chunks, and stored in a local database (ChromaDB).
One-time ingestion for local runs and vector retrieval.

System Architecture — Routing First



Safety Gate — Always First

Layer 1 · Keywords

- can't cope / go on
- no way forward
- end it / hurt myself
- suicid* / self-harm
- more than I can handle
- crisis · breakdown · desperate

Layer 2 · Embeddings

- Cosine similarity vs. escalate seeds
- Threshold: 0.70 (calibrated for recall)
- Catches novel phrasings keywords miss
- Shared model — no extra overhead
- Watch-level seeds at 0.60 threshold

Static, hand-authored response · No LLM call · Samaritans 116 123 · Shout 85258 · NHS 111

Tech Stack — Open Source, Low Cost, Reproducible

Embeddings	sentence-transformers/all-MiniLM-L6-v2
Vector Store	ChromaDB (persistent)
LLM	OpenAI GPT-5-mini
API	FastAPI + Pydantic v2
Frontend	Streamlit